A Brief Exploration of Helicopter Noise Reduction

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AS5370: Helicopter Aerodynamics

Technical Essay

Helicopter Noise Reduction

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Abstract

Helicopter noise reduction is one of the public-relation problem that causes issues while night-flying or expanding an airport. It is also one of the significant factor in military helicopters which can lead to a succesfull mission: long range propagation of helicopter with loud noise can alert an enemy to an incoming helicopter and provide them suffucient time to re-orient their defenses (the information is gathered by recognising accoustic signatures). This report makes an attempt to comprehend the possible sources of helicopter noise, methods of accoustic signature recognition and understand what techniques and methods, that can be applied to effectively reduce helicopter noise.

We will look into some terminology such as accoustic quiteing: process of making machinery quiter by damping vibrations to prevent them from reaching observer, We will understand phenomenon such as- Blade vortex interaction(BVI): which is an unsteady phenomenon of three dimensional nature, which occurs when a rotor blade passes within close proximity of the shed tip vortices from a previous blade and look into designs such Blue Edge blade design: An advanced design of rotor blade that reportedly achieves noise a reduction in noise emmision, vibrations and increases aerodynamic efficiency. We will then proceed to look into sources of noises such as rotor noise, anti-torque noise, engine noise. Also look into the previuos attempts made such as Eurocopter EC130: adoption of anti-torque device, and other models as well to understand the steps taken to reduce noise and effectively look into methods that can be adapted or modeified to create a better solution for our problem.

Helicopter noise sources and related noise generation mechanisms

- **Rotor noise**

i. **Loading noise –** Loading noise is an aerodynamic adverse effect due to the acceleration of the force distribution on the air around the rotor blade due to the blade passing through it, and is directed primarily below the rotor. In general, loading noise can include numerous types of blade loading. More over a rotating blade at non-zero angle of attack imposes rotating forces onto the surrounding air, causing blade loading noise. This sound generally propagates in a direction perpendicular to the plane of the rotor. Loading noise is also always occur, even in a hover condition.

ii. **Blade-vortex interaction noise –** Blade-vortex interaction (BVI) occurs when a rotor blade passes within a close proximity of the shed tip vortices from a previous blade. This causes a rapid, impulsive change in the loading on the blade resulting in the generation of highly directional impulsive loading noise. BVI noise can occur on either the advancing or retreating side of the rotor disk and its directivity is characterized by the precise orientation of the interaction. In general, advancing side BVI noise is directed down and forward while retreating-side BVIs cause noise that is directed down and rearward.

iii. **Thickness noise –** Thickness noise is caused by the blade periodically displacing air during each revolution. This sound propagates in the plane of the rotor. Thickness noise always occur, even in a hover condition.

iv. **High speed impulsive noise** – In level flight the blade's rotational speed adds to the flight speed to result in higher speeds on the advancing side, with the blade angle of attack at a minimum. On the retreating side the blade tip speed subtracts from the flight speed to cause locally minimal flow speeds with angle of attack at a maximum, at times even resulting in local flow separation (dynamic stall). Maximum speeds on the advancing side may cause the periodic appearance of aerodynamic shocks on the blade surface, resulting in high speed impulsive (HSI) noise. When these shocks delocalize from the rotor blade, they exhibit the long propagation distances and very high annoyance levels typical of shock waves. HSI noise is typically directed in the rotor plane forward of the helicopter, like thickness noise.

v. **Tail rotor noise** – While most noise from a helicopter is generated by the main rotor, the tail rotor is a significant source of noise for observers relatively close to the helicopter, where the higher-frequency noise of the tail rotor has not yet been attenuated by the atmosphere. Tail rotor noise is particularly annoying to the human listener due to its higher frequency (as compared to the main rotor) which places it directly in the band in which the human ear is most sensitive.

• Anti-torque noise

The noise mechanisms for the anti-torque system are basically identical to the rotor description in the paragraph above. However due to its position behind the main rotor, the anti-torque device can also be subject to non-uniform inflow caused by the main rotor wake. This leads to additional interaction noise phenomena. Alternative concepts like the NOTAR or the Fenestron system feature ducted rotors that have a somewhat different noise characteristic due to the shielding effect of the duct in the rotor rotational plane. In the special case of the NOTAR system the blower is located completely inside the tail boom. The air is guided from the blower through the tail boom and exits through slits in the tail boom thus generating the necessary anti-torque.

• Engine noise

The noise emitted by turboshaft engines is basically composed of the mainly rotational noise produced by the radial or axial compressors and turbine stages and broadband noise generated in the combustion chamber. Turboshaft engine exhaust noise has a broadband character and can become more prominent once the helicopter has passed overhead of the observer when rotor noise sources become less dominant.

Piston engines are typically used on smaller helicopters and can be one of the prominent noise sources for those aircraft. Exhaust noise typically dominates piston engine noise emissions and, for helicopters, most piston engine noise reduction has been focused on use of upturned exhausts, mufflers and resonators. Unsilenced exhaust noise is broadband with the highest levels at low frequencies.

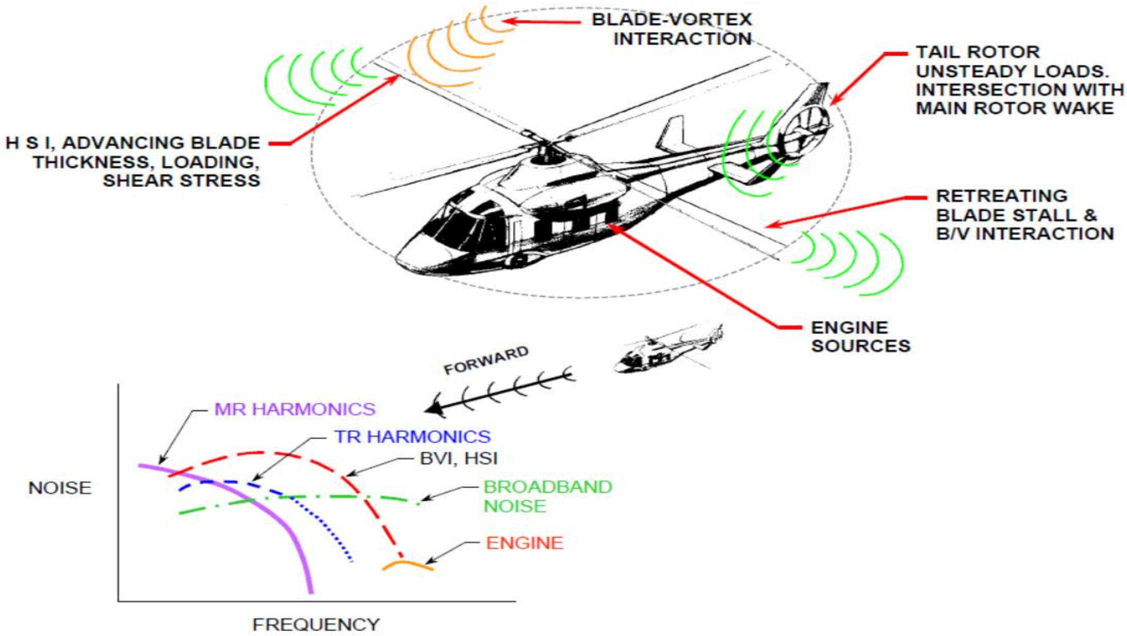


Figure- Helicopter noise propagates from numerous sources

Acoustic Quieting

Acoustic quieting is the process of making machinery quieter by damping vibrations to prevent them from reaching the observer. Machinery vibrates, causing sound waves in air, hydroacoustic waves in water, and mechanical stresses in solid matter. Quieting is achieved by absorbing the vibrational energy or minimizing the source of the vibration. It may also be redirected away from the observer.

Acoustic quieting might consider...

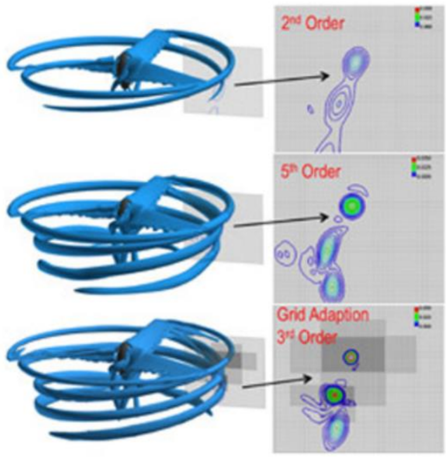
- Noise generation: by limiting the noise at its source,
- Sympathetic vibrations: by acoustic decoupling,
- Resonations: by acoustic damping or changing the size of the resonator,
- Sound transmissions: by reducing transmission using many methods (depending whether the transmission is through air, liquid, or solid), or
- Sound reflections: by limiting the reflection using many methods, e.g. by using acoustic absorption (deadening) materials, trapping the sound, opening a "window" to let sound out, etc.

By observing the source of noise production, an engineer can focus on reducing it's noise by dampening the material, isolating it, trying to remove any barrier that's causing such vibrations in the atmosphere.

Blade-vortex interaction (BVI)

A blade vortex interaction (BVI) is an unsteady phenomenon of three-dimensional nature, which occurs when a rotor blade passes within a close proximity of the shed tip vortices from a previous blade.

It represents an important topic of investigation in rotorcrafts.



The BVIs are predicted by practical testing as well as analytical method. Example: HART I (Higher Harmonic Control Aero-acoustic Rotor Test I)- In this test, a 40% scaled BO-105 rotor model along with a fuselage is used, a range of sophisticated measurement techniques are introduced to measure the noise level, blade surface pressure, tip vortices, and structural moments.

Current analytic method involves free wake methods using Full-Potential Equations, Euler Equations and Navier-stokes Equations. The simulations are performed via Computational fluid dynamics(CFD). The method results in larger values than real ones due to ignorance of wake effects. To make the solution more accurate, researchers are trying to use multiple methods and merge the data to get more accurate solutions. Discrete vortex models were also introduced to make the numerical results much realistic.

Blue Edge blade

Blue Edge main rotor blade provides a passive reduction in noise levels, using a double-swept shape that is very different from normal blades. The aim of this design is to reduce the noise generated by so-called blade-vortex interactions (BVI), which occur when a blade impacts a vortex, created at the tip of the blade of any helicopter. It is also help to reduce vibration and increased aerodynamics efficiency. The structure decreases the clamor produced by cutting edge vortex collaboration (BVI), the trademark throbbing sound made when the tip of a rotor edge hits the vortex shed by the tip of the previous sharp edge. Blue Edge tip shape gives more opportunity for the communication between the edge and vortex, weakening the noise.

Blue edge is an advanced design of rotor blade developed and produced by multinational helicopter manufacturer Airbus helicopter. The Blue Edge rotor was originally developed at ONERA and the German Aerospace Center under the auspices of the ERATO program. A five-blade Blue Edge main rotor has been flying since July 2007 on an EC155 testbed, logging 75 flight hours and demonstrating noise reductions of 3 to 4 dB, as well as very good performance of the blade.

According to aerospace researcher Yves Delrieux, the design of the Blue Edge rotor blade was defined using tools developed by ONERA from wind tunnel testing to model aerodynamic performance; these tools were claimed to have achieved a reduction in noise emissions and vibration as well as greater aerodynamic efficiency over conventional modes

Figure- Blue edge rotor

(a) Blue edge blade (b) Conventional blade

Figure- Comparison of blades BVI

Overview of technology programs and research initiatives for noise reduction

The overall situation with respect to major noise technology research initiatives worldwide is summarized in below Figure. It covers a 15-year period (2001-2015), providing an evolutionary perspective from the previous helicopter noise research survey performed before 2000.

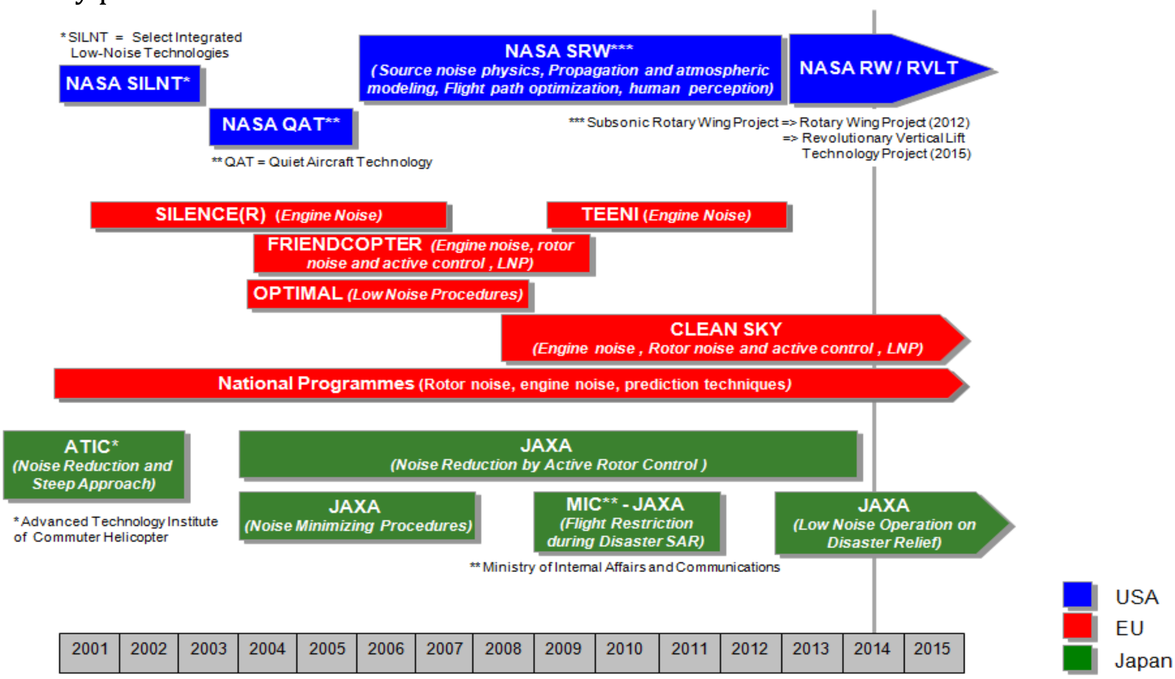


Figure- Major Initiatives in Helicopter Noise Research Since 2000

The major initiatives (eg. USA, EU, and Japan) are represented and a summary of some of these research programs is provided in the following sections.

1. UNITED STATES

- **Subsonic Rotary Wing (SRW)**
  In 2012, the Subsonic Rotary Wing Project reorganized and was renamed the Rotary Wing Project, continuing the emphasis on performance, efficiency and noise research. Noise research was addressed in the areas of:

- Cabin Noise
  - Structural / Acoustic Modeling for Interior Noise
  - Noise Mitigation and Design Concepts
- Human Response to Rotorcraft Noise
- Rotorcraft Aeroacoustics
  - Source Noise Physics
  - Flight Aeroacoustics
  - Rotorcraft Noise Mitigation Science

- **NASA Revolutionary Vertical Lift Technology (RVLT)**
  In 2015, the NASA Revolutionary Vertical Lift Technology (RVLT) Project was formed in a major re-structuring of the NASA Aeronautics Research Mission Directorate. RVLT develops and validates tools, technologies and concepts to overcome key barriers for rotary wing vehicles. RVLT research primarily focuses on vehicles that require mature technology solutions in the 2020-2030 timeframe. The scope of the noise research continues to encompass source noise physics, interior noise modeling, and flight aeroacoustics. Modeling and analysis validated with experimental data has been the emphasis of the research in all areas.

2. EUROPEAN UNION

- **TEENI (Turboshaft Engine Exhaust Noise Identification)**

The major deliverables of TEENI are:

i. A comprehensive full-scale engine test database, the first of its kind, with extensive internal measurements and far field instrumentation.
ii. A noise breakdown realised out of a panel of original signal processing methods which have been developed during the project, using internal measurements to understand the origin of noise measured in the far field.

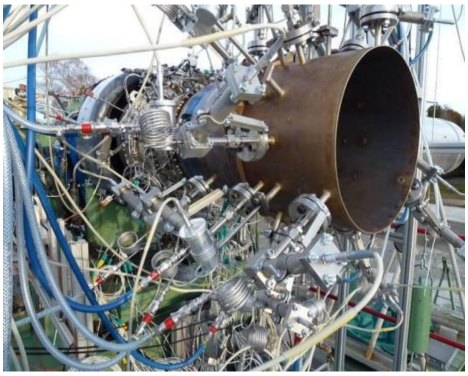


Figure- Engine experimental setup

- **FRIENDCOPTER (Integration of technologies in support of a passenger and environmentally friendly helicopter)**

Main objectives of the research programme were:

i. Helicopter acoustic footprint areas reduced between 30% and 50% depending on the flight condition,
ii. Fuel consumption reduction up to 6% for high speed flights,
iii. Cabin noise levels near 75 dB(A) similar to airliner cabins for cruise flight,
iv. Cabin vibrations below 0.05 g corresponding to jet smooth ride comfort for the same flight regime.

- **CLEAN SKY Joint Technology Initiative**

The general Clean Sky objectives and the GRC top-level objectives are to:

i. reduce CO_2 emission by 25 to 40% per mission (for rotorcraft powered respectively by turboshaft or diesel engines);

ii. reduce the noise perceived on ground by 10 EPN dB or halving the noise footprint area by 50%;

iii. ensure full compliance with the REACH directive which protects human health and environment from harmful chemical substances.

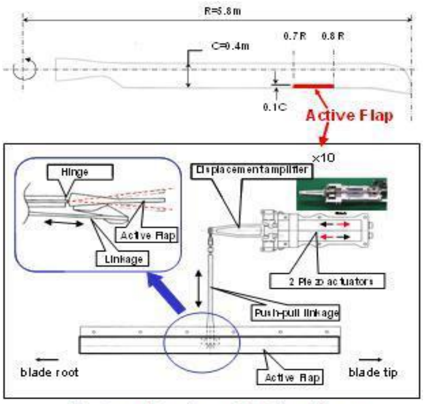
- **ABC (Active Blade Control)**
  The ABC project concerned rotors equipped with trailing edge flaps. Following numerical optimization studies a model rotor was built and tested in the S1-Modane wind tunnel. The test proved a significant reduction of vibration and BVI noise. The research for the ABC rotor was a collaborative German French project by DLR and ONERA.
  Figure- Active flap rotor in S1-Modane wind tunnel

3. JAPAN

- **Research Program of ATIC**
  ATIC (Advanced Technology Institute of Commuter-helicopter, Ltd.) was established in 1994 in order to research and develop the technologies to reduce the helicopter external noise for public acceptance and to enhance the flight safety in a 7-year research program. The final goal for the noise reduction area was to develop the technologies that can realize the helicopter external noise level at least 10 EPN dB less than the current ICAO noise limit.

Model Rotor Testing at DNW

- **Noise Reduction by Active Rotor Control**
  JAXA started the research and development program of the full-scale active flap. As the first step, the conceptual study of the active flap system was carried out and the full-scale on-board active flap system was developed which demonstrated its operability by the bench test. In the second step, a transonic wind tunnel test is performed on the condition to cover simulated several flight patterns such as landing/approach, hover, assumed maximum cruising speed of a helicopter in order to demonstrate the proper operability of the active flap system on more realistic environment.



System Design of Active Flap

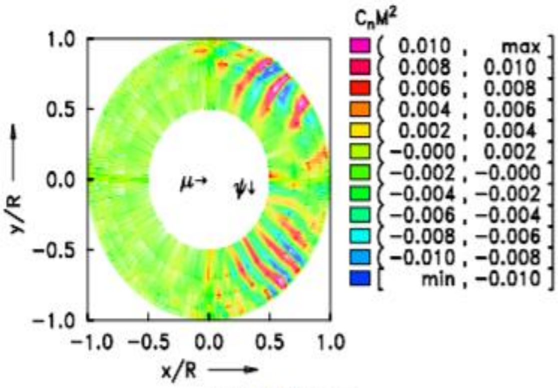
Methods to reduce noise

i. **Spoilers** - By attaching spoiler in the helicopter main rotor blades and tail rotors reduce the noise and silence the operation. The spoiler is made of supple, resilient, and durable non-ferrous material, such as rubber, a rubber like material, plastic, rubber-impregnated with thin steel mesh, canvas reinforced rubber, rubber composites, and other durable materials that do not prematurely fatigue. The spoiler attached at the trailing edges of blades have free edges exhibits a non-repeating pattern of flow which collectively break up the vortex formed by blade rotation by providing clean air for each successive blade. The blade efficiency is also increased by using spoiler. The drag and power requirement would certainly increase due to introducing spoilers. This method can effectively decrease BVI noise.

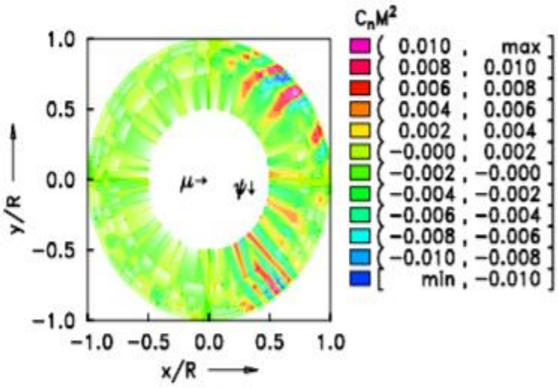
ii. **Number of blades** - Increasing number of blades does reduce the noise generation but it also increases the need of higher power requirement.

iii. **Tail noise reduction** - Focusing on tail rotor's noise, the reduction in noise can be performed by changing the design of the helicopter such that the interference of main rotor wake is minimalistic. Another idea is to remove the tail completely but that effectively works for small helicopters, larger helicopter will need counter-moment mechanism which are either not feasible in terms of mechanical complexity or due to increasing the noises.

iv. **Blade designs** - Design can be made that are capable to perform better. One such example is blue edge rotor blade. Such blades are proven to be noise efficient as well as aerodynamic efficient as well. The figure represents the BVI location and intensity and given aerodynamic parameters. The top figure represents simpler blade and the bottom one represents blue edge-based design blades. It is visible that the bottom figure shows lower amount of BVI intensity, therefore increasing the noise efficiency.



(a) 7AD rotor



v. **Pitch** - The stealth or noise efficiency can be achieved by increasing the angle of attack of the blades (pitch), such that the blades does not experience stall angle (so effectively there's a cap to maximum angle of attack).

Summary

Understanding the helicopter noise sources and their behaviour as well as the significance of each corresponding sources such as- Loading noise, Blade-votex interaction noise, thickness noise, high speed impulsive noise, tail rotor noise and the engine noise. Understanding the process of quitening of machenery, the theoritical methods such Blade -vortex interaction(BVI), Computer fluid dynamics(CFD) with the help of Euler, Navier-stoke, Full-potential equation and assertions such as Blue edge blade design that can be used to understand the noise reduction. then proceeding to look into the past progress in the field of accoustic quietening for helicopters and understand the patters of progress- made so far.Briefly looking into technology programs and research initiatives performed by United states- Subsonic Rotary Wing (SRW), NASA Revolutionary vertical lift technology (RLVT), European Union- Helicopter noise and Vibration reduction (HELINOVI), Turboshaft Engine Exhaust Noise Identification (TEENI), Integration of technologies in support of a passenger and environmentally friendly helicopter (FRIENDCOPTER), CLEAN SKY Joint Technology Initiative, Active Blade Control (ABC), and Japan- Research Program of ATIC, Noise Reduction by Active Rotor Control. And suggesting some methods to help further in reduction of noises by factors such as spoilers, number of blades, tail noise reduction and elimination, blade design and pitch. And the conclusion of information that is gathered and accumulated.

Conclusion

Currently researchers are developing more accurate performance simulations with the help of modern comuters, computational fluid dynamics and descrete vortex models. We looked into the past researches and innitiatives performed by some major countries and observed the experiments as well as the results. We break down the data into important aspects of methods that can be effectively considered in reducing noise for a given helicopter. The idea is to keep the changes minimalistic yet try reduce the noise as much as possible. We also looked into the behavior of noise reduction as a correalation to the blade design and it's intereference with the presence of tail.

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